PROJECT DESIGN PHASE – II

Technology Stack (Architecture & Tools)

Project	Global Fuel Prices and Taxation Trends (2020–2026)
Prepared for	Anushka Dubey
Team ID	To be provided
Project period	2020–2026 (data coverage)
Platform	Tableau Public

Technology Stack Overview

The project uses a file-based data source and Tableau Public for business intelligence and visualisation. The CSV dataset is connected to Tableau, where the data fields are used to build calculated views, KPI cards, filters and an interactive dashboard. No separate web application, database server or backend service is identified as part of the implemented project.

Technology Components

S. No.	Component	Role in the project	Technology used	Purpose / justification
1	Data source	Stores the quarterly fuel-price records used for analysis.	CSV dataset	Portable tabular source containing country, region, year, quarter, fuel type, price, estimated tax share and income group.
2	Data connection	Makes the dataset available to the visualisation environment.	Tableau Public – text/CSV connection	Connects the source fields to the Tableau workbook.
3	Data preparation	Supports checking and using the required fields for analysis.	Tableau data pane and field settings	Enables field selection, data-type review and use of dimensions and measures in views.
4	Analysis & calculations	Supports aggregations and comparisons used in charts and KPI cards.	Tableau calculations and aggregations	Used for measures such as average fuel price and estimated tax-share comparisons, as configured in the workbook.
5	Visualisation	Builds the analytical sheets and dashboard.	Tableau Public	Creates Fuel Price Trends, Regional Comparison, Country Ranking and Tax Analysis views.
6	User interaction	Lets users explore subsets of the data.	Tableau filters and dashboard interactions	Filters include year, region and fuel type.
7	Publishing / sharing	Makes the finished dashboard accessible for review and demonstration.	Tableau Public	Provides a shareable public dashboard/workbook link when published.

High-level Solution Architecture

Layer	Project implementation	Output
Data layer	CSV dataset: 7,056 quarterly records covering 84 countries, 7 regions, 3 fuel types and 2020–2026.	Structured fuel-price and estimated-tax fields.
Connection & preparation	Connect the CSV to Tableau Public and verify field names, data types and measures.	Usable dimensions and measures.
Analysis layer	Use Tableau aggregations, comparisons and calculated fields where configured.	Trend, regional, country and tax comparisons.
Visualisation layer	Create four worksheets and KPI cards, then arrange them in a dashboard.	Interactive Global Fuel Prices Dashboard.

Layer	Project implementation	Output
User layer	Apply year, region and fuel-type filters to explore the dashboard.	Focused comparisons and visual insights.

Implementation Notes & Limitations

• The dataset used for this project is simulated. Results are suitable for demonstrating dashboard design and comparative analysis, not for reporting official current fuel prices or tax rates.
• The project is implemented as a Tableau Public dashboard; a separate custom front end, API, SQL database or cloud data pipeline is not listed as an implemented component.
• Any future refresh should use a verified, documented data source and preserve consistent units and definitions.

Technology names above are limited to the tools and project setup established in the available project details; unconfirmed tools have not been added.